

Skill Session 1

TASK 1

```
jella_pragathi@LAPTOP-RQQGT555:~$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
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warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

git version 2.53.0
GNU Make 4.4.1
Built for x86_64-pc-linux-gnu
Copyright (C) 1988-2023 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <https://gnu.org/licenses/gpl.html>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
jella_pragathi@LAPTOP-RQQGT555:~$ mkdir -p ~/ossp_project
jella_pragathi@LAPTOP-RQQGT555:~$ cd ~/ossp_project
jella_pragathi@LAPTOP-RQQGT555:~/ossp_project$ mkdir src include build
jella_pragathi@LAPTOP-RQQGT555:~/ossp_project$ touch src/main.c
touch Makefile
jella_pragathi@LAPTOP-RQQGT555:~/ossp_project$ ls
Makefile  build  include  src
jella_pragathi@LAPTOP-RQQGT555:~/ossp_project$ tree
.
├── Makefile
├── build
├── include
└── src
    └── main.c

4 directories, 2 files
jella_pragathi@LAPTOP-RQQGT555:~/ossp_project$ nano src/main.c
jella_pragathi@LAPTOP-RQQGT555:~/ossp_project$ cat src/main.c
#include <stdio.h>
int main() {
    printf("OSSP Project Started Successfully!\n");
    return 0;
}
```

```
jella_pragathi@LAPTOP-RQQGT555:~/ossp_project$ nano Makefile
jella_pragathi@LAPTOP-RQQGT555:~/ossp_project$ cat Makefile
CC = gcc
CFLAGS = -Wall -Wextra -g
TARGET = build/main
SRC = src/main.c
all:
    $(CC) $(CFLAGS) $(SRC) -o $(TARGET)
clean:
    rm -f $(TARGET)
run: all
    ./$(TARGET)
jella_pragathi@LAPTOP-RQQGT555:~/ossp_project$ make
gcc -Wall -Wextra -g src/main.c -o build/main
jella_pragathi@LAPTOP-RQQGT555:~/ossp_project$ ls build
main
jella_pragathi@LAPTOP-RQQGT555:~/ossp_project$ ./build/main
OSSP Project Started Successfully!
jella_pragathi@LAPTOP-RQQGT555:~/ossp_project$ |
```

```
GNU nano 8.7.1 src/main.c
#include <stdio.h>

int main() {
    printf("OSSP Project Started Successfully!\n");
    return 0;
}
```

TASK 2


```

jella_pragathi@LAPTOP-RQQT555:~/ossp_project$ ./ossp_project
-bash: ./ossp_project: Is a directory
jella_pragathi@LAPTOP-RQQT555:~/ossp_project$ nano src/myshell.c
jella_pragathi@LAPTOP-RQQT555:~/ossp_project$ gcc -Wall -Wextra -g src/myshell.c -o build/myshell
jella_pragathi@LAPTOP-RQQT555:~/ossp_project$ ./build/myshell
Mini Shell Started
Type 'exit' to quit.

myshell> hello
You entered: hello
myshell> exit
Exiting shell...
jella_pragathi@LAPTOP-RQQT555:~/ossp_project$ |

```

```

GNU nano 8.7.1 src/myshell.c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

#define MAX_INPUT 100

int main() {
    char input[MAX_INPUT];

    printf("Mini Shell Started\n");
    printf("Type 'exit' to quit.\n\n");

    while (1) {
        printf("myshell> ");
        fflush(stdout);

        if (fgets(input, sizeof(input), stdin) == NULL) {
            break;
        }

        input[strcspn(input, "\n")] = '\0';

        if (strcmp(input, "exit") == 0) {
            printf("Exiting shell...\n");
            break;
        }

        if (strlen(input) == 0) {
            continue;
        }

        printf("You entered: %s\n", input);
    }

    return 0;
}

```

TASK 2

```
Exiting shell...
jella_pragathi@LAPTOP-RQQGT555:~/ossp_project$ nano src/keyboard_shell.c
jella_pragathi@LAPTOP-RQQGT555:~/ossp_project$ gcc -Wall -Wextra -g src/keyboard_shell.c -o build/keyboard_shell
jella_pragathi@LAPTOP-RQQGT555:~/ossp_project$ ./build/keyboard_shell
Keyboard Input Shell
Type commands and press Enter.
Use Backspace to edit.
Type 'exit' to quit.

myshell> hello
Command entered: hello
myshell> exit
Exiting shell...
jella_pragathi@LAPTOP-RQQGT555:~/ossp_project$ tree
.
├── Makefile
├── build
│   ├── keyboard_shell
│   ├── main
│   ├── myshell
│   └── process_demo
├── include
└── src
    ├── keyboard_shell.c
    ├── main.c
    ├── myshell.c
    └── process_demo.c

4 directories, 9 files
jella_pragathi@LAPTOP-RQQGT555:~/ossp_project$ |
```

```
#include <stdio.h>
#include <unistd.h>
#include <termios.h>
#include <string.h>

#define MAX_INPUT 100

void enable_raw_mode(struct termios *original) {
    struct termios raw;

    tcgetattr(STDIN_FILENO, original);
    raw = *original;

    raw.c_lflag &= ~(ICANON | ECHO);

    tcsetattr(STDIN_FILENO, TCSAFLUSH, &raw);
}

void disable_raw_mode(struct termios *original) {
    tcsetattr(STDIN_FILENO, TCSAFLUSH, original);
}

int main() {
    struct termios original;
    char buffer[MAX_INPUT];
    int position = 0;
    char ch;

    printf("Keyboard Input Shell\n");
    printf("Type commands and press Enter.\n");
    printf("Use Backspace to edit.\n");
    printf("Type 'exit' to quit.\n\n");

    enable_raw_mode(&original);

    while (1) {
        printf("myshell> ");
        fflush(stdout);

        position = 0;

        while (1) {
            if (read(STDIN_FILENO, &ch, 1) <= 0) {
                break;
            }
        }
    }
}
```

PRACTICE QUESTIONS

Simple commands:

```
jella_pragathi@LAPTOP-RQQGT555:~$ pwd
/home/jella_pragathi
jella_pragathi@LAPTOP-RQQGT555:~$ ls
OSSP
jella_pragathi@LAPTOP-RQQGT555:~$ date
Fri Aug 7 08:15:01 UTC 2026
jella_pragathi@LAPTOP-RQQGT555:~$ whoami
jella_pragathi
jella_pragathi@LAPTOP-RQQGT555:~$ hostname
LAPTOP-RQQGT555
jella_pragathi@LAPTOP-RQQGT555:~$ uname
Linux
jella_pragathi@LAPTOP-RQQGT555:~$ uname -r
6.18.33.2-microsoft-standard-WSL2
jella_pragathi@LAPTOP-RQQGT555:~$ id
uid=1000(jella_pragathi) gid=1000(jella_pragathi) groups=1000(jella_pragathi),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),100(users)
jella_pragathi@LAPTOP-RQQGT555:~$ tty
/dev/pts/0jella_pragathi@LAPTOP-RQQGT555:~$ uptime
08:15:57 up 2 min, 1 user, load average: 0.02, 0.03, 0.00
jella_pragathi@LAPTOP-RQQGT555:~$ users
jella_pragathi@LAPTOP-RQQGT555:~$ who
jella_pragathi@LAPTOP-RQQGT555:~$ w
08:16:23 up 3 min, 1 user, load average: 0.01, 0.02, 0.00
USER      TTY      FROM            LOGIN@   IDLE   JCPU   PCPU   WHAT
jella_pr pts/1        -          08:13    3:02   0.04s  0.04s -bash
jella_pragathi@LAPTOP-RQQGT555:~$ groups
jella_pragathi adm cdrom sudo dip plugdev users
```

```
jella_pragathi@LAPTOP-RQQGT555:~$ env
SHELL=/bin/bash
WSL2_GUI_APPS_ENABLED=1
WSL_DISTRO_NAME=Ubuntu
NAME=LAPTOP-RQQGT555
PWD=/home/jella_pragathi
LOGNAME=jella_pragathi
HOME=/home/jella_pragathi
LANG=C.UTF-8
WSL_INTEROP=/run/WSL/327_interop
LS_COLORS=rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33;01:cd=40;33;01:or=40;31;01:mi=00:su=37;41:sg=30;43:ca=00:tw=30;42:ow=34;42:st=37;40:ex=01;32*:tar=01;31*:taz=01;31*:arc=01;31*:arj=01;31*:taz=01;31*:lha=01;31*:lzh=01;31*:lzh=01;31*:lzma=01;31*:tlz=01;31*:txz=01;31*:tzo=01;31*:t7z=01;31*:zip=01;31*:z=01;31*:dz=01;31*:gz=01;31*:lrz=01;31*:lzo=01;31*:xz=01;31*:zst=01;31*:tzt=01;31*:bz2=01;31*:bz=01;31*:tbz=01;31*:tbz2=01;31*:tz=01;31*:deb=01;31*:rpm=01;31*:jar=01;31*:war=01;31*:ear=01;31*:sar=01;31*:rar=01;31*:alz=01;31*:ace=01;31*:zoo=01;31*:cpi=01;31*:7z=01;31*:rz=01;31*:cab=01;31*:wim=01;31*:swm=01;31*:dwm=01;31*:esd=01;31*:avif=01;35*:jpg=01;35*:jpeg=01;35*:mjpg=01;35*:mjpeg=01;35*:gif=01;35*:bmp=01;35*:pbm=01;35*:pgm=01;35*:ppm=01;35*:tga=01;35*:xbm=01;35*:xpm=01;35*:tif=01;35*:tiff=01;35*:png=01;35*:svg=01;35*:svgz=01;35*:mng=01;35*:pex=01;35*:mov=01;35*:mpg=01;35*:mpeg=01;35*:m2v=01;35*:mkv=01;35*:webm=01;35*:webp=01;35*:ogm=01;35*:mp4=01;35*:m4v=01;35*:mp4v=01;35*:vob=01;35*:qt=01;35*:nuv=01;35*:wmv=01;35*:asf=01;35*:rm=01;35*:rmvb=01;35*:flc=01;35*:avi=01;35*:fli=01;35*:flv=01;35*:gl=01;35*:dl=01;35*:xcf=01;35*:xwd=01;35*:yuv=01;35*:cgm=01;35*:emf=01;35*:ogv=01;35*:ogx=01;35*:aac=00;36*:au=00;36*:flac=00;36*:m4a=00;36*:mid=00;36*:midi=00;36*:mka=00;36*:mp3=00;36*:mpc=00;36*:ogg=00;36*:ra=00;36*:wav=00;36*:oga=00;36*:opus=00;36*:spx=00;36*:xspf=00;36*:#=#=00;90*:bak=00;90*:old=00;90*:orig=00;90*:part=00;90*:rej=00;90*:swp=00;90*:tmp=00;90*:dpkg-dist=00;90*:dpkg-old=00;90*:ucf-dist=00;90*:ucf-new=00;90*:ucf-old=00;90*:rpmnew=00;90*:rpmorig=00;90*:rpmsave=00;90:
WAYLAND_DISPLAY=wayland-0
LESSCLOSE=/usr/bin/lesspipe %s %s
TERM=xterm-256color
LESSOPEN=| /usr/bin/lesspipe %s
USER=jella_pragathi
DISPLAY=:0
SHLVL=1
XDG_RUNTIME_DIR=/run/user/1000
WSLENV=
XDG_DATA_DIRS=/usr/local/share:/usr/share:/var/lib/snapd/desktop
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/WindowsApps/MicrosoftCorporationII.WindowsSubsystemForLinux.2.7.11.0_x64_8wekyb3d8bbwe:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/windows/system32:/mnt/c/windows/System32/Wbem:/mnt/c/windows/System32/WindowsPowerShell/v1.0:/mnt/c/windows/System32/WindowsPowerShell/v1.0:/mnt/c/windows/System32/WindowsPowerShell/v1.0:/mnt/c/Users/jella_pragathi/Downloads/gcc-15.1.0/bin:/mnt/c/Program Files/Git/cmd:/mnt/c/Program Files/nodejs:/mnt/c/Program Files/MySQL/MySQL Shell 8.0/bin:/mnt/c/Users/jella_pragathi/AppData/Local/Programs/Python/Python314/Script s:/mnt/c/Users/jella_pragathi/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/jella_pragathi/AppData/Roaming/npm:/snap/bin
DBUS_SESSION_BUS_ADDRESS=unix:path=/run/user/1000/bus
HOSTTYPE=x86_64
```

```
jella_pragathi@LAPTOP-RQQT555:~$ printenv
SHELL=/bin/bash
WSL2_GUI_APPS_ENABLED=1
WSL_DISTRO_NAME=Ubuntu
NAME=LAPTOP-RQQT555
PWD=/home/jella_pragathi
LOGNAME=jella_pragathi
HOME=/home/jella_pragathi
LANG=C.UTF-8
WSL_INTEROP=/run/WSL/327.interop
LS_COLORS=rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33;01:cd=40;33;01:or=40;31;01:mi=00:su=37;41:sg=30;43:ca=00:tw=30;42:ow=34;42:st=37;44:ex=01;32:*.tar=01;31:*.tgz=01;31:*.arc=01;31:*.arj=01;31:*.taz=01;31:*.lha=01;31:*.lz4=01;31:*.lzh=01;31:*.lzma=01;31:*.tlz=01;31:*.txz=01;31:*.tzo=01;31:*.v7z=01;31:*.zip=01;31:*.z=01;31:*.dz=01;31:*.gz=01;31:*.lrz=01;31:*.lzo=01;31:*.xz=01;31:*.zst=01;31:*.tzt=01;31:*.bz2=01;31:*.bz=01;31:*.tbz=01;31:*.tbz2=01;31:*.t=01;31:*.deb=01;31:*.rpm=01;31:*.jar=01;31:*.war=01;31:*.ear=01;31:*.sar=01;31:*.rar=01;31:*.alz=01;31:*.ace=01;31:*.zoo=01;31:*.cpio=01;31:*.7z=01;31:*.rz=01;31:*.cab=01;31:*.wim=01;31:*.swm=01;31:*.esd=01;31:*.avif=01;35:*.jpg=01;35:*.jpeg=01;35:*.mjpg=01;35:*.mjpeg=01;35:*.gif=01;35:*.bmp=01;35:*.pbm=01;35:*.pgm=01;35:*.ppm=01;35:*.tga=01;35:*.xbm=01;35:*.xpm=01;35:*.tif=01;35:*.tiff=01;35:*.png=01;35:*.svg=01;35:*.svgz=01;35:*.mng=01;35:*.pcx=01;35:*.pcx=01;35:*.mov=01;35:*.mpeg=01;35:*.mp2=01;35:*.mkv=01;35:*.webm=01;35:*.webp=01;35:*.ogm=01;35:*.ogg=01;35:*.mp4=01;35:*.m4v=01;35:*.mpjv=01;35:*.vob=01;35:*.qt=01;35:*.nuv=01;35:*.mvm=01;35:*.asf=01;35:*.rm=01;35:*.rmvb=01;35:*.flc=01;35:*.avi=01;35:*.flv=01;35:*.gl=01;35:*.dl=01;35:*.xcf=01;35:*.xwd=01;35:*.yuv=01;35:*.cgm=01;35:*.emf=01;35:*.ogv=01;35:*.ogx=01;35:*.aac=00;36:*.au=00;36:*.flac=00;36:*.m4a=00;36:*.mid=00;36:*.midi=00;36:*.mka=00;36:*.mp3=00;36:*.mpc=00;36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=00;36:*.opus=00;36:*.spx=00;36:*.xspf=00;36:*.xspf=00;36:*.xspf=00;36:*.xspf=00;36:*.bak=00;90:*.old=00;90:*.orig=00;90:*.part=00;90:*.rej=00;90:*.swp=00;90:*.tmp=00;90:*.dpkg-dist=00;90:*.dpkg-old=00;90:*.ucf-dist=00;90:*.ucf-new=00;90:*.ucf-old=00;90:*.rpmnew=00;90:*.rpmorig=00;90:*.rpmsave=00;90:
WAYLAND_DISPLAY=wayland-0
LESSCLOSE=/usr/bin/lesspipe %s %s
TERM=xterm-256color
LESSOPEN=| /usr/bin/lesspipe %s
USER=jella_pragathi
DISPLAY=:0
SHLVL=1
XDG_RUNTIME_DIR=/run/user/1000
WSLENV=
XDG_DATA_DIRS=/usr/local/share:/usr/share:/var/lib/snapd/desktop
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/WindowsApps/MicrosoftCorporationII.WindowsSubsystemForLinux.2.7.11.0_x64_8wekyb3d8bbwe:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/windows/system32:/mnt/c/windows:/mnt/c/windows/System32/Wbem:/mnt/c/windows/System32/WindowsPowerShell/v1.0:/mnt/c/windows/System32/OpenSSH:/mnt/c/Users/jella_pragathi/Downloads/gcc-15.1.0/bin:/mnt/c/Program Files/Git/cmd:/mnt/c/Users/jella_pragathi/AppData/Local/Programs/Python/Python314:/mnt/c/Users/jella_pragathi/AppData/Local/Programs/Python/Python314/Scripts:/mnt/c/Users/jella_pragathi/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/jella_pragathi/AppData/Roaming/npm:/snap/bin
DBUS_SESSION_BUS_ADDRESS=unix:path=/run/user/1000/bus
HOSTTYPE=x86_64
```

```
jella_pragathi@LAPTOP-RQQT555:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
none            3.8G   0  3.8G   0% /usr/lib/modules/6.18.33.2-microsoft-standard-WSL2
none            3.8G  4.0K  3.8G   1% /mnt/wsl
drivers          129G  111G   19G  87% /usr/lib/wsl/drivers
/dev/sdd        1007G  1.8G  954G   1% /
none            3.8G   36K  3.8G   1% /mnt/wslg
none            3.8G   0  3.8G   0% /usr/lib/wsl/lib
rootfs          3.8G  2.8M  3.8G   1% /init
none            3.8G  552K  3.8G   1% /run
none            3.8G   0  3.8G   0% /run/lock
none            3.8G   0  3.8G   0% /run/shm
none            3.8G   80K  3.8G   1% /mnt/wslg/versions.txt
none            3.8G   80K  3.8G   1% /mnt/wslg/doc
C:\             129G  111G   19G  87% /mnt/c
D:\             348G  315M  347G   1% /mnt/d
none            1.0M   0  1.0M   0% /run/credentials/systemd-journald.service
tmpfs           3.8G   0  3.8G   0% /tmp
none            1.0M   0  1.0M   0% /run/credentials/systemd-resolved.service
none            1.0M   0  1.0M   0% /run/credentials/getty@tty1.service
tmpfs           766M  12K  766M   1% /run/user/1000
jella_pragathi@LAPTOP-RQQT555:~$ free -h
               total        used        free      shared  buff/cache   available
Mem:           7.5Gi        573Mi        6.4Gi        3.5Mi        685Mi        6.9Gi
Swap:          2.0Gi           0B          2.0Gi
```

```
jella_pragathi@LAPTOP-RQQT555:~$ ps -ef
UID      PID     PPID    C  STIME TTY          TIME CMD
root         1         0  0  08:13 ?        00:00:00 /sbin/init
root         2         1  0  08:13 hvc0     00:00:00 /init
root         7         2  0  08:13 hvc0     00:00:00 plan9 --control-socket 7
root        47         1  0  08:13 ?        00:00:00 /usr/lib/systemd/systemd
systemd+    82         1  0  08:13 ?        00:00:00 /usr/lib/systemd/systemd
root        90         1  0  08:13 ?        00:00:00 /usr/lib/systemd/systemd
root       111         1  0  08:13 ?        00:00:00 /bin/sh /usr/lib/systemd
root       112         1  0  08:13 ?        00:00:00 /usr/sbin/cron -f -P
message+   113         1  0  08:13 ?        00:00:00 @dbus-daemon --system --
root       152         1  0  08:13 ?        00:00:00 /usr/bin/python3 /usr/bi
root       174         1  0  08:13 ?        00:00:00 /usr/lib/systemd/systemd
root       177         1  0  08:13 ?        00:00:00 /usr/libexec/wsl-pro-ser
syslog     193         1  0  08:13 ?        00:00:00 /usr/sbin/rsyslogd -n -i
root      210         1  0  08:13 tty1     00:00:00 /usr/sbin/agetty --nores
_chrony    227       111  0  08:13 ?        00:00:00 /usr/sbin/chronyd -n -F
_chrony    244       227  0  08:13 ?        00:00:00 /usr/sbin/chronyd -n -F
root      269         1  0  08:13 ?        00:00:00 /usr/bin/python3 /usr/sh
root      326         2  0  08:13 ?        00:00:00 /init
root      327       326  0  08:13 ?        00:00:00 /init
jella_p+   332       327  0  08:13 pts/0    00:00:00 -bash
root      334         2  0  08:13 ?        00:00:00 login -- jella_pragathi
jella_p+   388         1  0  08:13 ?        00:00:00 /usr/lib/systemd/systemd
jella_p+   391       388  0  08:13 ?        00:00:00 (sd-pam)
jella_p+   420       334  0  08:13 pts/1    00:00:00 -bash
jella_p+   670       332  0  08:17 pts/0    00:00:00 ps -ef
```

```
jella_pragathi@LAPTOP-RQQT555:~$ top
top - 08:20:42 up 7 min, 1 user, load average: 0.09, 0.05, 0.01
Tasks: 25 total, 1 running, 24 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 99.9 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7651.2 total, 6446.0 free, 608.1 used, 749.3 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7043.1 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	24476	15528	11620	S	0.0	0.2	0:00.84	systemd
2	root	20	0	3180	2208	2072	S	0.0	0.0	0:00.01	init-systemd(Ub
7	root	20	0	3180	2044	1940	S	0.0	0.0	0:00.00	init
47	root	19	-1	50424	16732	15464	S	0.0	0.2	0:00.20	systemd-journal
82	systemd+	20	0	22540	14640	12020	S	0.0	0.2	0:00.08	systemd-resolve
90	root	20	0	35444	12396	9224	S	0.0	0.2	0:00.38	systemd-udev
111	root	20	0	2888	1952	1820	S	0.0	0.0	0:00.05	chronyd-starter
112	root	20	0	4472	3168	2924	S	0.0	0.0	0:00.00	cron
113	message+	20	0	8800	5408	4720	S	0.0	0.1	0:00.05	dbus-daemon
152	root	20	0	44092	29704	14564	S	0.0	0.4	0:00.30	networkd-dispat
174	root	20	0	18436	9448	8200	S	0.0	0.1	0:00.11	systemd-logind
177	root	20	0	1792300	15380	12552	S	0.0	0.2	0:00.12	wsl-pro-service
193	syslog	20	0	220548	5940	4624	S	0.0	0.1	0:00.09	rsyslogd
210	root	20	0	5492	2724	2540	S	0.0	0.0	0:00.01	agetty
227	_chrony	20	0	24252	11780	7236	S	0.0	0.2	0:00.09	chronyd
244	_chrony	20	0	12092	2404	1760	S	0.0	0.0	0:00.00	chronyd
269	root	20	0	123456	32632	18024	S	0.0	0.4	0:00.20	unattended-upgr
326	root	20	0	3184	1112	980	S	0.0	0.0	0:00.00	SessionLeader
327	root	20	0	3200	1248	1108	S	0.0	0.0	0:00.02	Relay(332)
332	jella_p+	20	0	6268	5536	3864	S	0.0	0.1	0:00.13	bash
334	root	20	0	8648	5604	4760	S	0.0	0.1	0:00.01	login
388	jella_p+	20	0	22508	13336	10892	S	0.0	0.2	0:00.15	systemd
391	jella_p+	20	0	22912	4088	2084	S	0.0	0.1	0:00.00	(sd-pam)
420	jella_p+	20	0	6136	5400	3852	S	0.0	0.1	0:00.04	bash
680	jella_p+	20	0	8172	6032	3920	R	0.0	0.1	0:00.11	top

```
jella_pragathi@LAPTOP-RQQT555:~$ vmstat
procs -----memory-----swap-- ----io---- -system-- -----cpu-----
r b swpd free buff cache si so bi bo in cs us sy id wa st gu
0 0 0 6598512 49852 718208 0 0 974 20 435 0 0 0 100 0 0 0

jella_pragathi@LAPTOP-RQQT555:~$ iostat
Command 'iostat' not found, but can be installed with:
sudo apt install sysstat
jella_pragathi@LAPTOP-RQQT555:~$ sudo apt update
sudo apt install sysstat -y
[sudo: authenticate] Password:
Hit:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:3 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Hit:4 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:5 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [447 kB]
Get:6 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [371 kB]
Get:7 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [89.6 kB]
Get:8 http://archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [111 kB]
Get:9 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [84.7 kB]
Get:10 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [32.8 kB]
Get:11 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [235 kB]
Get:12 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.1 kB]
Get:13 http://archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [76.9 kB]
Get:14 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [321 kB]
Get:15 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [166 kB]
Get:16 http://archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [331 kB]
Get:17 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [61.2 kB]
Get:18 http://archive.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [63.0 kB]
Get:19 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse amd64 Packages [11.4 kB]
Get:20 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse Translation-en [3032 B]
Fetched 2723 kB in 8s (333 kB/s)
110 packages can be upgraded. Run 'apt list --upgradable' to see them.
Installing:
  sysstat

Suggested packages:
  isag

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 110
  Download size: 517 kB
```

```

2 erms invpcid rdseed adx smap clflushopt clwb
sha_ni xsaveopt xsavec xgetbv1 xsaves avx_vnni
vnm i umip waitpkg gfni vaes vpclmulqdq rdpid mo
vdiri movdir64b fsr m md_clear serialize ibt flu
sh_lld arch_capabilities

Virtualization features:
  Virtualization: VT-x
  Hypervisor vendor: Microsoft
  Virtualization type: full
Caches (sum of all):
  L1d: 432 KiB (9 instances)
  L1i: 576 KiB (9 instances)
  L2: 18 MiB (9 instances)
  L3: 18 MiB (1 instance)
NUMA:
  NUMA node(s): 1
  NUMA node0 CPU(s): 0-17
Vulnerabilities:
  Gather data sampling: Not affected
  Ghostwrite: Not affected
  Indirect target selection: Not affected
  Itlb multihit: Not affected
  L1tf: Not affected
  Mds: Not affected
  Meltdown: Not affected
  Mmio stale data: Not affected
  Old microcode: Not affected
  Reg file data sampling: Not affected
  Retbleed: Mitigation; Enhanced IBRS
  Spec rstack overflow: Not affected
  Spec store bypass: Mitigation; Speculative Store Bypass disabled via prctl
  Spectre v1: Mitigation; usercopy/swapgs barriers and __user pointer sanitization
  Spectre v2: Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSE-IBRS SW sequence; BHI BHI_DIS_S
  Srbds: Not affected
  Tsa: Not affected
  Tsx async abort: Not affected
  Vmscape: Not affected

```

```

jella_pragathi@LAPTOP-RQQGT555:~$ lsblk
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda    8:0    0 356.9M  1 disk
sdb    8:16   0 159.4M  1 disk
sdc    8:32   0    2G    0 disk [SWAP]
sdd    8:48   0    1T    0 disk /mnt/wslg/distro
/

```

```

jella_pragathi@LAPTOP-RQQGT555:~$ mount
none on /usr/lib/modules/6.18.33.2-microsoft-standard-WSL2 type overlay (rw,nosuid,nodev,noatime,lowerdir=/modules,upperdir=/lib/modules/6.18.33.2-microsoft-standard-WSL2/rw/upper,workdir=/lib/modules/6.18.33.2-microsoft-standard-WSL2/rw/work,uid=on)
none on /mnt/wsl type tmpfs (rw,relatime)
drivers on /usr/lib/wsl/drivers type 9p (ro,nosuid,nodev,noatime,aname=drivers,fmask=222,dmask=222,cache=0x5,access=client,msize=65536,trans=fd,rfd=8,wfd=8)
/dev/sdd on / type ext4 (rw,relatime,discard,errors=remount-ro,data=ordered)
none on /mnt/wslg type tmpfs (rw,relatime)
/dev/sdd on /mnt/wslg/distro type ext4 (ro,relatime,discard,errors=remount-ro,data=ordered)
none on /usr/lib/wsl/lib type overlay (rw,nosuid,nodev,noatime,lowerdir=/gpu_lib_packaged:/gpu_lib_inbox,upperdir=/gpu_lib/rw/upper,workdir=/gpu_lib/rw/work,uid=on)
rootfs on /init type rootfs (ro,size=3911632k,nr_inodes=977908)
none on /dev type devtmpfs (rw,nosuid,relatime,size=3911632k,nr_inodes=977908,mode=755)
sysfs on /sys type sysfs (rw,nosuid,nodev,noexec,noatime)
proc on /proc type proc (rw,nosuid,nodev,noexec,noatime)
devpts on /dev/pts type devpts (rw,nosuid,noexec,noatime,gid=5,mode=620,ptmxmode=000)
none on /run type tmpfs (rw,nosuid,nodev,mode=755)
none on /run/lock type tmpfs (rw,nosuid,nodev,noexec,noatime)
none on /run/shm type tmpfs (rw,nosuid,nodev,noatime)
none on /dev/shm type tmpfs (rw,nosuid,nodev,noatime)
none on /run/user type tmpfs (rw,nosuid,nodev,noexec,noatime,mode=755)
binfmt_misc on /proc/sys/fs/binfmt_misc type binfmt_misc (rw,relatime)
cgroup2 on /sys/fs/cgroup type cgroup2 (rw,nosuid,nodev,noexec,relatime,nsdelegate)
none on /mnt/wslg/versions.txt type overlay (rw,relatime,lowerdir=/systemvhd,upperdir=/system/rw/upper,workdir=/system/rw/work,uid=on)
none on /mnt/wslg/doc type overlay (rw,relatime,lowerdir=/systemvhd,upperdir=/system/rw/upper,workdir=/system/rw/work,uid=on)
none on /tmp/.X11-unix type tmpfs (ro,relatime)
C:\ on /mnt/c type 9p (rw,noatime,aname=drvfs;path=C:\;uid=1000;gid=1000;symlinkroot=/mnt/,cache=0x5,access=client,msize=65536,trans=fd,rfd=6,wfd=6)
D:\ on /mnt/d type 9p (rw,noatime,aname=drvfs;path=D:\;uid=1000;gid=1000;symlinkroot=/mnt/,cache=0x5,access=client,msize=65536,trans=fd,rfd=6,wfd=6)
none on /run/user type tmpfs (rw,relatime)
none on /run/credentials/systemd-journald.service type tmpfs (ro,nosuid,nodev,noexec,relatime,nosymfollow,size=1024k,nr_inodes=1024,mode=700,noswap)
mqueue on /dev/mqueue type mqueue (rw,nosuid,nodev,noexec,relatime)
tmpfs on /tmp type tmpfs (rw,nosuid,nodev,size=3917392k,nr_inodes=1048576)
tracefs on /sys/kernel/tracing type tracefs (rw,nosuid,nodev,noexec,relatime)
hugetlbfs on /dev/hugepages type hugetlbfs (rw,nosuid,nodev,relatime,pagesize=2M)
debugfs on /sys/kernel/debug type debugfs (rw,nosuid,nodev,noexec,relatime)
fusectl on /sys/fs/fuse/connections type fusectl (rw,nosuid,nodev,noexec,relatime)
configfs on /sys/kernel/config type configfs (rw,nosuid,nodev,noexec,relatime)
none on /run/credentials/systemd-resolved.service type tmpfs (ro,nosuid,nodev,noexec,relatime,nosymfollow,size=1024k,nr_inodes=1024,mode=700,noswap)
none on /tmp/.X11-unix type tmpfs (ro,relatime)
none on /run/credentials/getty@tty1.service type tmpfs (ro,nosuid,nodev,noexec,relatime,nosymfollow,size=1024k,nr_inodes=1024,mode=700,noswap)
tmpfs on /run/user/1000 type tmpfs (rw,nosuid,nodev,relatime,size=783476k,nr_inodes=195869,mode=700,uid=1000,gid=1000)
tmpfs on /mnt/wslg/run/user/1000 type tmpfs (rw,nosuid,nodev,relatime,size=783476k,nr_inodes=195869,mode=700,uid=1000,gid=1000)

```

```
jella_pragathi@LAPTOP-RQQT555:~$ history
1  pwd
2  clear
3  pwd
4  history
5  date
6  clear
7  whoami
8  cal
9  pwd
10 whoami
11 hostname
12 uname-a
13 mkdir OSSP
14 cd OSSP
15 git init
16 mkdir src include obj bin docs tests scripts assets
17 touch .gitignore README.md LICENSE Makefile
18 touch src/{main.c,shell.c,parser.c,executor.c,builtin.c}
19 touch include/{shell.h,parser.h,executor.h,builtin.h}
20 touch bin/my_shell
21 touch docs/{design.md,report.pdf}
22 touch tests/{test_parser.c,test_executor.c}
23 touch scripts/run.sh
24 chmod +x scripts/run.sh
25 touch assets/architecture.png
26 tree
27 sudo apt install tree
28 mkdir OSSP
29 cd OSSP
30 git init
31 mkdir src include obj bin docs tests scripts assets
32 touch .gitignore README.md LICENSE Makefile
33 touch src/{main.c,shell.c,parser.c,executor.c,builtin.c}
34 touch include/{shell.h,parser.h,executor.h,builtin.h}
35 touch bin/my_shell
36 touch docs/{design.md,report.pdf}
37 touch tests/{test_parser.c,test_executor.c}
38 touch scripts/run.sh
39 chmod +x scripts/run.sh
40 touch assets/architecture.png
```

```
62 ps
63 ps -ef
64 top
65 pwd
66 history
67 pwd
68 ls
69 date
70 whoami
71 hostname
72 uname
73 uname -r
74 id
75 tty
76 uptime
77 users
78 who
79 w
80 groups
81 env
82 printenv
83 df -h
84 free -h
85 ps -ef
86 top
87 q
88 vmstat
89 iostat
90 sudo apt update
91 sudo apt install sysstat -y
92 iostat
93 lscpu
94 lsblk
95 mount
96 history
jella_pragathi@LAPTOP-RQQT555:~$ |
```

```
jella_pragathi@LAPTOP-RQQT555:~$ echo "Hello World"
Hello World
jella_pragathi@LAPTOP-RQQT555:~$ sudo apt install bc -y
[sudo: authenticate] Password:
bc is already the newest version (1.07.1-4build1).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 110
jella_pragathi@LAPTOP-RQQT555:~$ bc
bc 1.07.1
Copyright 1991-1994, 1997, 1998, 2000, 2004, 2006, 2008, 2012-2017 Free Software Foundation, Inc.
This is free software with ABSOLUTELY NO WARRANTY.
For details type `warranty'.
^C
(interrupt) use quit to exit.
5+10
20*3
100/4
quit
15
60
25
jella_pragathi@LAPTOP-RQQT555:~$ factor 100
100: 2 2 5 5
jella_pragathi@LAPTOP-RQQT555:~$ sleep 5
jella_pragathi@LAPTOP-RQQT555:~$
```